FEATURE

Looking ahead at our future and how our relation with AI would evolve with Intel!

5 things Intel revealed about our AI future

Intel had some really interesting insights to share about our AI future!

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Intel seems to have rediscovered its mojo, if the company's strategic direction revealed by CEO Pat Gelsinger at Intel Innovation 2023 is anything to go by. Apart from trying to supercharge its chip offerings for both the consumer retail and datacenter business, Intel is also attempting to reverse its slide and give the likes of NVIDIA and TSMC a proper run for their money in the more lucrative AI and foundry businesses.

Here are five key things we learned about the new and improved Intel from its annual innovation conference, areas where Intel is focusing extensively to give them a competitive edge now and in the near future.

1 The AI PC era is almost here

The very first slide of Pat Gelsinger's keynote presentation said "bringing AI everywhere," which also includes the good old PC (desktops and laptops). At a time when everyone and their grandparents are talking about AI, Intel emphasized its strategy to embed AI into every aspect of its hardware and software platform – that means Meteor Lake on PCs to Sapphire Rapids on servers.

In particular, the 14th gen Intel Core Ultra (codenamed Meteor Lake) will debut some of the company's key improvements to semiconductors, including with respect to AI. With product launch confirmed for December 14, 2023, not only is Intel Core Ultra based on Intel 4 process technology (7nm node) but it's Intel's first chip built using EUV, chiplets and enhanced Foveros 3D packaging, allowing it to deliver 2X more performance at roughly half the power draw compared to Intel 7 (10nm node) process technology, according to early claims. It's also going to have a brand new neural processing unit or NPU engine for dedicated AI tasks, which will offer a greater ability to switch between onboard GPU for sustained AI-infused tasks and delegate lesser resource intensive AI workloads to the CPU.

From a developer's perspective, there can be no doubt that Intel is laying the foundation for increased use of AI applications on its upcoming chips for both consumers and businesses alike – unlocked undoubtedly by a solid software stack on top. Similar to what Apple is doing with the iPhone and Mac (with its Neural Engine) or Qualcomm's Snapdragon AI on Android phones. Not surprisingly, it's being called the iPhone moment of AI for PC, as it will allow third-party developers and OEMs to tap into AI and machine learning use cases on the PC like never before.

2 Rise of Hybrid AI on the edge

With increased infusion of AI-on-chip, Intel foresees an acceleration of on-device AI workloads, especially on PCs, laptops and other areas of the software-defined edge – everything from fast food kiosks to digital electric meters, from providing a safer healthcare experience to making factories more integrated and beyond.

Whether it's interacting with a fast food restaurant's chatbot to help select the best food items under 500 calories or reducing a doctor's or healthcare worker's clerical work inside hospitals, client devices at the edge of the network will become smarter, more intelligent and have the ability to run nimble AI models or Generative AI tasks locally on-device, according to Sachin Katti, senior vice president and general manager of the Network and Edge Group (NEX) at Intel Corporation.

This is happening for two major reasons, first and foremost because running everything AI-related in the cloud or datacenters alone isn't exactly cheap. More AI integration into devices at the edge will